



11

Geometry

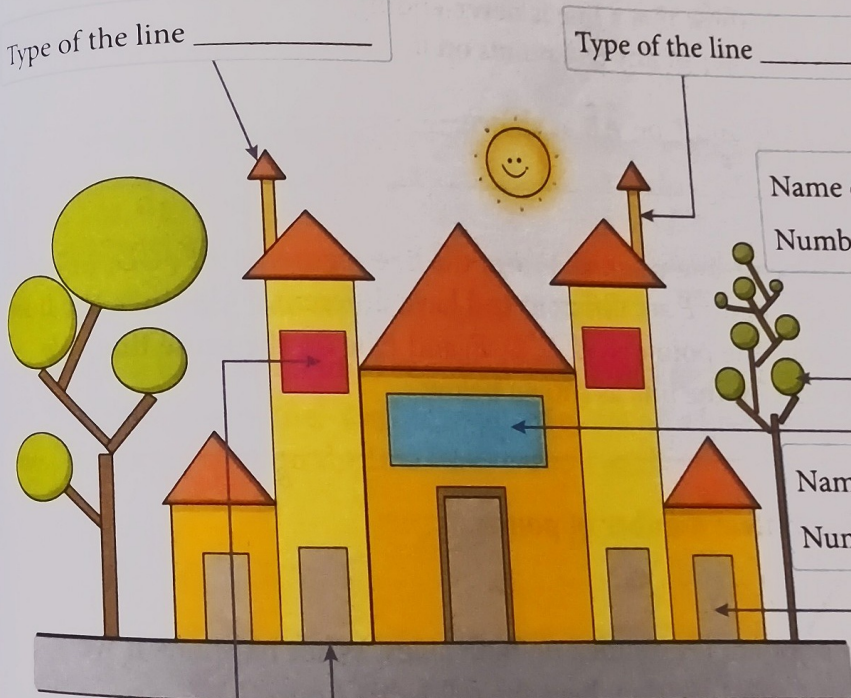
Tune In

Go to MyInspire for  



Suruchi studies in grade 3 and she has made a beautiful picture.

- Identify the names of the shapes in the given picture. Write the names of the shapes and the total number of sides of the shape you see in the picture.
- Identify and write the type of lines—horizontal, vertical, and slant.



Type of the line _____

Type of the line _____

Name of the shape _____
Number of sides _____

Name of the shape _____
Number of sides _____

Type of the line _____

Name of the shape _____
Number of sides _____

Name of the shape _____
Number of sides _____

BASIC GEOMETRICAL CONCEPTS

Point

A point is a position in space. It has no length, breadth, or height. It is represented by a dot (.).

A capital letter is used to name a point.

• M
Point M

Line Segment

A line segment is formed when two points are joined using a ruler. The two points are called the end points of the line segment and are used to name the line segment.

A line segment has a definite length.

$\overline{XY}$ or $\overline{YX}$
Line segment XY

Line

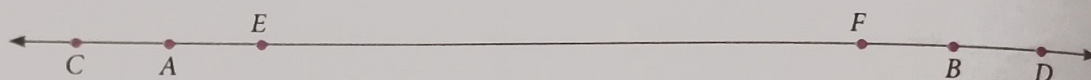
When a line segment is extended on both sides, it forms a line.

The arrow heads on both ends show that a line is never-ending.

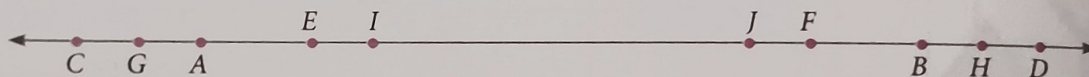
A line is named using a small letter or any two points on it.

$\overleftrightarrow{AB}$
Line m or $\overleftrightarrow{AB}$

Now let us mark points C, D, E, and F on $\overleftrightarrow{AB}$ as shown.



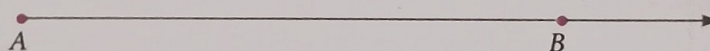
Can you see that you get the same line when you form the line segments $\overline{AB}$, $\overline{CD}$, $\overline{EF}$, and extend them? Though $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$ are different and have different endpoints, the lines $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{EF}$ are the same. The points A, B, C, D, E, and F lie on the same line. We can mark any number of points on the line as shown below.



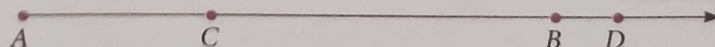
Thus, a line is made up of countless number of points.

Ray

A line is formed when a line segment is extended on both sides. What happens if we extend the segment only on one side? We then have the following figure:



The above figure has one fixed endpoint A. But it extends beyond B. So it is not a line segment. Since it has one endpoint, it is not a line. This figure is called a ray. The above ray is identified as $\overrightarrow{AB}$. The arrowhead is only in the direction of B. Instead of B, any point on the ray can be used to represent the ray.



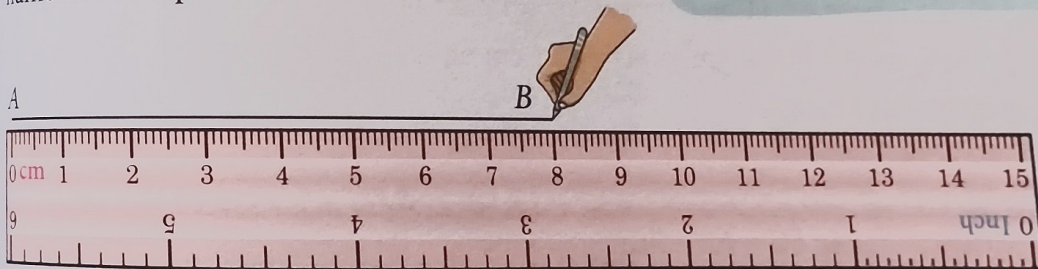
The rays $\vec{AC}$ and $\vec{AD}$ are the same as $\vec{AB}$. Like a line, a ray is also made up of a countless number of points. The difference is that while a line does not have any endpoint, a ray does have one.

CONSTRUCTION OF LINE SEGMENTS

Let us learn to construct line segments.

To Draw a Line Segment AB of Length 8 cm

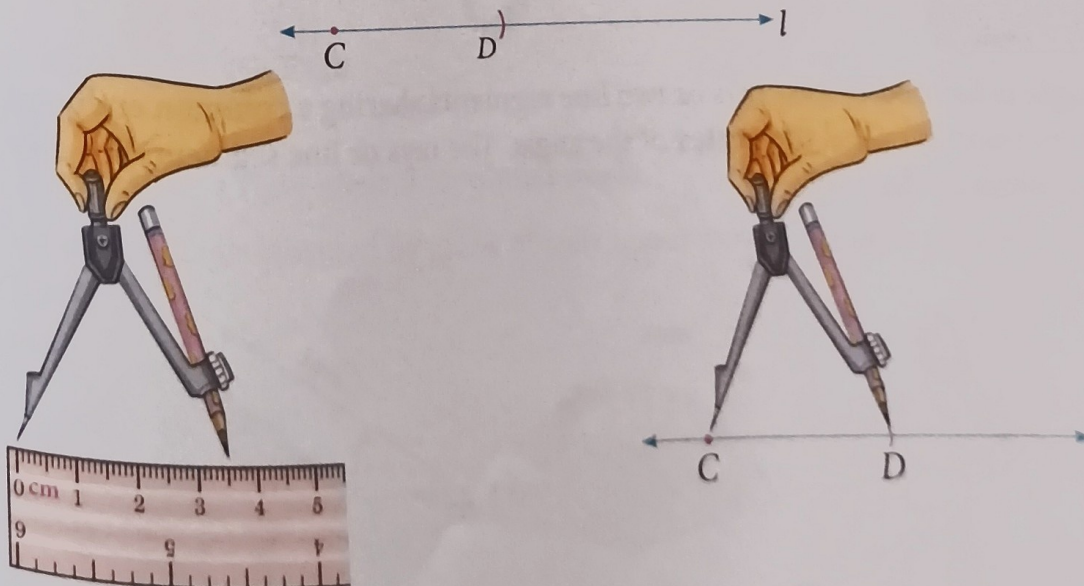
Place the ruler on the page of your notebook. Hold it firmly. Mark two points, one at the 0-cm mark of the ruler and the other at the 8-cm mark. Using a sharp pencil, join the two points by drawing along the straight edge of the ruler to obtain the line segment. Remove the ruler and name the two points as A and B .



AB is the required line segment of length 8 cm.

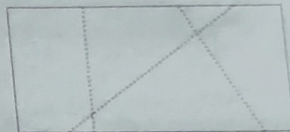
To cut off a line segment of required length on a given line

Let ' l ' be the given line. You are required to cut off a line segment CD of length 3.5 cm on line l . Mark a point on the line l and name it as C .



Let's Try! 1

Take a rectangular sheet of paper 9 cm by 4 cm.



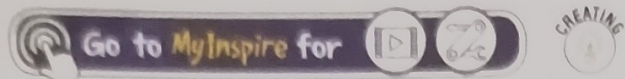
Fold the sheet in three different ways as shown and open it out. Measure the creases formed.

1. Each one of them is a line segment. Why?
2. Are they of the same length?
3. Which one is the shortest?
4. What is its length?

Fix a sharp pencil in the compass and stretch it on a ruler so that the metal tip is at the 0 cm mark and the pencil tip is at the 3.5 cm mark. Now, carefully place the metal point at C and using the pencil tip mark the point ' D ' on the line l . Thus CD is the required line segment of length 3.5 cm.

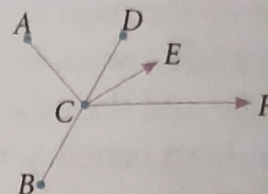
Let's Try! 2

1. Draw a line segment of 6 cm.
2. Draw a line segment of 4.5 cm.



Exercise 11A

1. Identify the rays and line segments in the given figure.
2. Draw line segments of the following lengths.
 - a. 3.7 cm
 - b. 4.8 cm
 - c. 6.6 cm
 - d. 10.4 cm
3. Draw a line ' m ' and cut off a line segment $AB = 6.2$ cm on it.
4. Draw a line ' n '. Cut off a line segment XY of 5.5 cm on it.

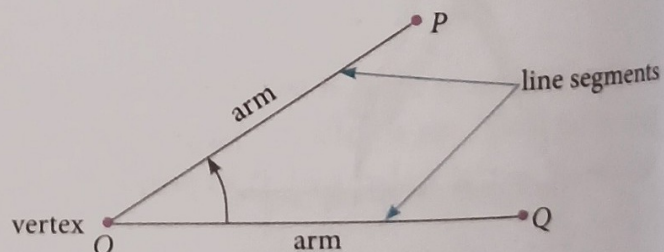
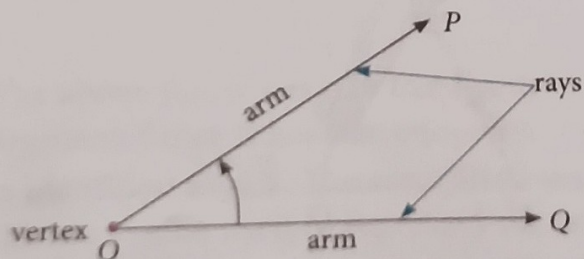


Application Based Questions

5. Draw a line ' q '. Cut off two line segments $EF = 4.6$ cm and $FG = 5.4$ cm in the same direction. Measure EG . Is $EG = EF + FG$?
6. Draw a line ' l '. Cut off a line segment $PQ = 8$ cm on it. Cut off another line segment $QR = 4.4$ cm in the opposite direction. Measure PR . What do you find?

ANGLES

An **angle** is formed by two rays or two line segments sharing a common endpoint. The common point is called the **vertex** of the angle. The rays or line segments are called the **sides** or **arms** of the angle.



$\angle$ is a symbol used to denote an angle. The angle formed at vertex O is denoted by $\angle O$ or $\angle POQ$. Similarly, the angle formed at vertex Z is denoted by $\angle Z$ or $\angle XZY$.

Key Point

The vertex at which an angle is formed must be written in the middle of the name. For instance, the vertex of $\angle ABC$ is at B.

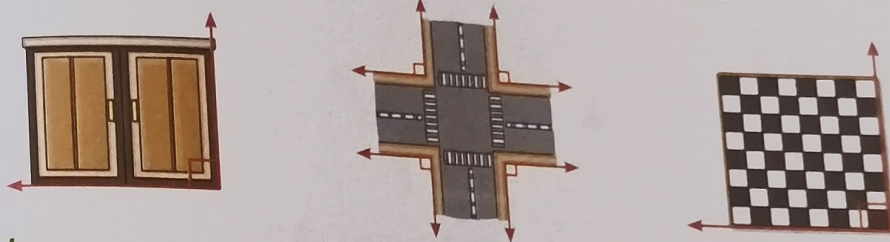
Types of Angles

Angles are classified based on their measures. Different types of angles are formed when two arms of an angle meet in different ways.

Right Angle

An angle formed by a vertical line and horizontal line is called a **right angle**.

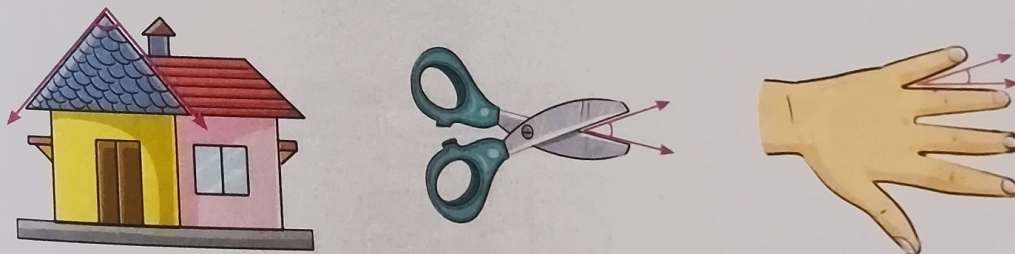
You can see right angles formed by many objects in our surrounding as shown in the given pictures.



Acute Angle

An angle less than a right angle is an **acute angle**.

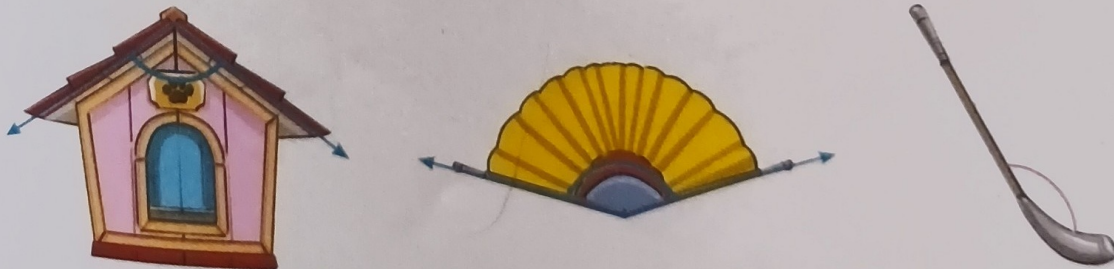
You can see acute angles formed by many objects in our surrounding as shown in the given pictures.



Obtuse Angle

An angle greater than a right angle is an **obtuse angle**.

You can see obtuse angles formed by many objects in our surrounding as shown in the given pictures.



Straight Angle

An angle formed by joining two right angles is **straight angle**.

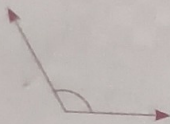
You can see straight angles formed by many objects in our surrounding as shown in the given pictures.



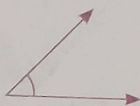
Exercise 11B

1. Name the angles.

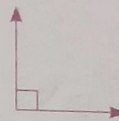
a.



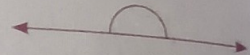
b.



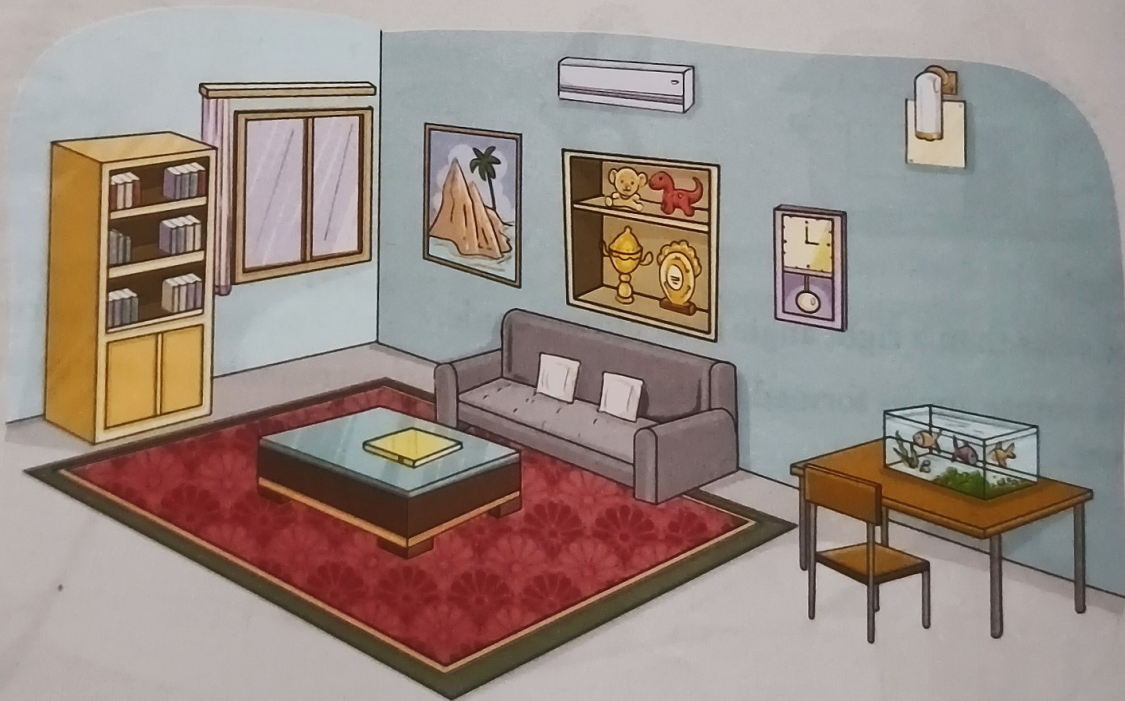
c.



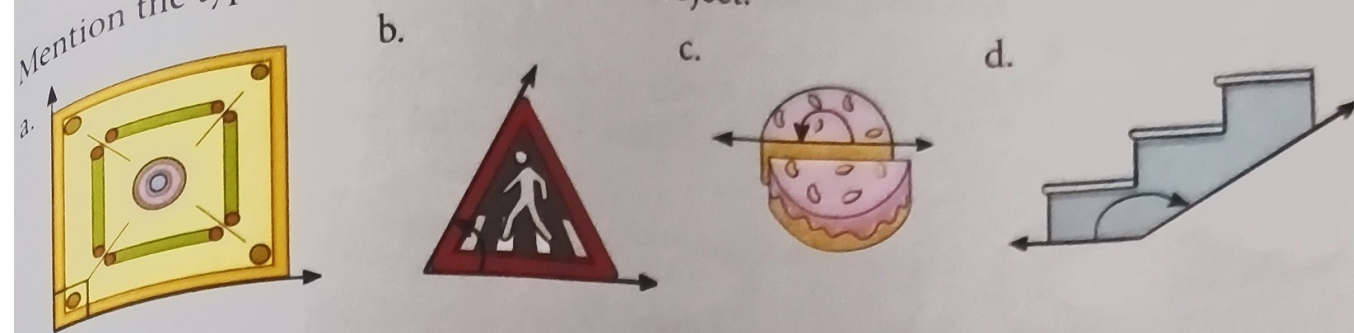
d.



2. Become a right-angle detective. Mark 10 different objects in the given picture which have right angles.

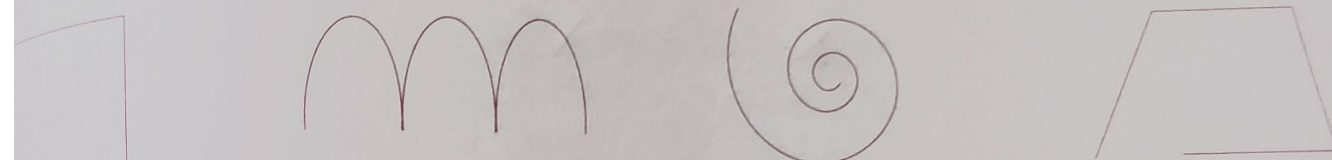


Mention the type of angle denoted in each object.



OPEN AND CLOSED FIGURES

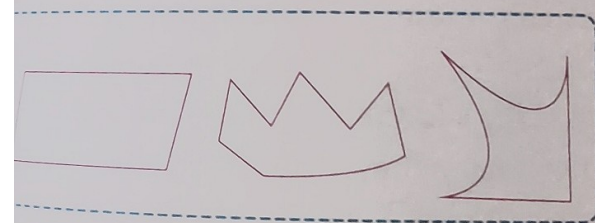
A shape that begins and ends at two different points is called an **open figure**. The figures given below are open figures.



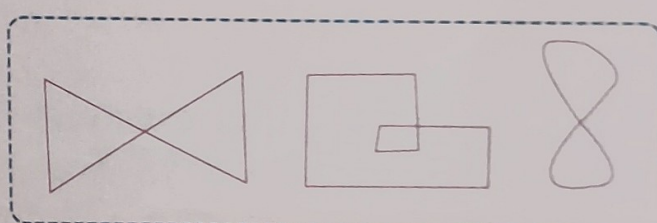
A shape that begins and ends at the same point is called a **closed figure**. The figures given below are closed figures.



A closed figure that does not cross itself is called a **simple closed figure**.



Simple closed figures



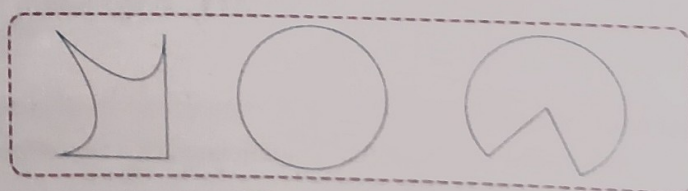
Not simple closed figures

Polygons

Simple closed figures made of line segments are called **polygons**.



Polygons

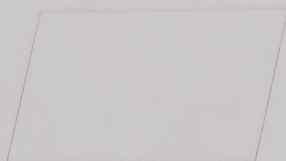


Not polygons

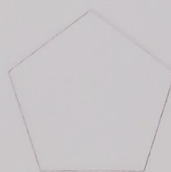
A polygon is named according to the number of sides.



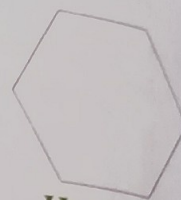
Triangle
3 sides



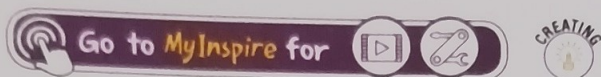
Quadrilateral
4 sides



Pentagon
5 sides



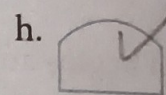
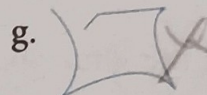
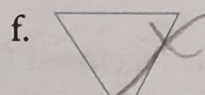
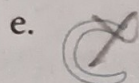
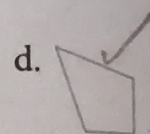
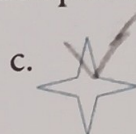
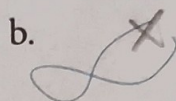
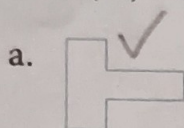
Hexagon
6 sides



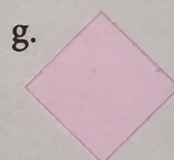
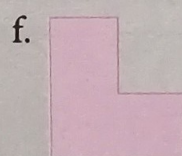
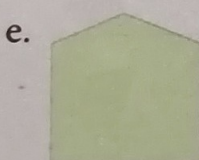
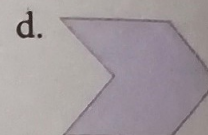
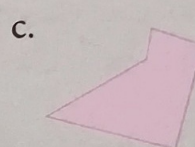
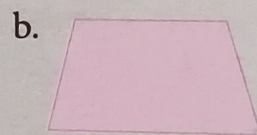
Exercise 11C

Go to MyInspire and attempt questions 1 and 2.

3. Tick (✓) the closed figures and cross (✗) the open figures.



4. Name the polygons according to the number of sides.



Application Based Questions

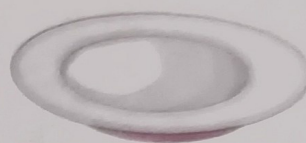
5. Trace the base of a dice. Is it a polygon? If yes, name it according to the number of sides.

Teacher's Note

Explain to the students that a quadrilateral is a closed four-sided figure and that rectangles and squares are special quadrilaterals.

CIRCLE

You will require any of the following objects: carom striker, round bottle cap, bangle, round button, round dinner plate, round sharpener, coin etc.



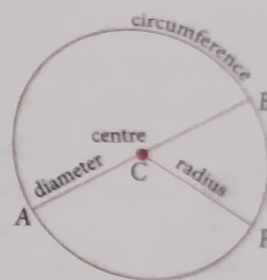
Place each of these objects on the page of your notebook and draw their outlines. What do you observe about these shapes?

They are all simple closed figures that do not have sides or corners. These are examples of circles.

Parts of a Circle

In the given circle, point C is the **centre**. The line segment AB has its end points on the boundary of the circle and it passes through the centre. It is a **diameter** of the circle.

CP is a line segment, one endpoint of which is the centre and the other lies on the circle. Such a line segment is called a **radius** of the circle.



Now, use your ruler and draw one more diameter naming its end points D and E.

Also, draw another radius CX. Use your ruler and measure the following.

1. diameters AB and DE. Is $AB = DE$?
2. radii (plural of radius) CP and CX. Is $CP = CX$?
3. What is the relation between AB and CP? You will find that $2 \times \text{radius} = \text{diameter}$.
4. Check whether any radius or diameter you draw has the same property—
 $2 \times \text{radius} = \text{diameter}$.

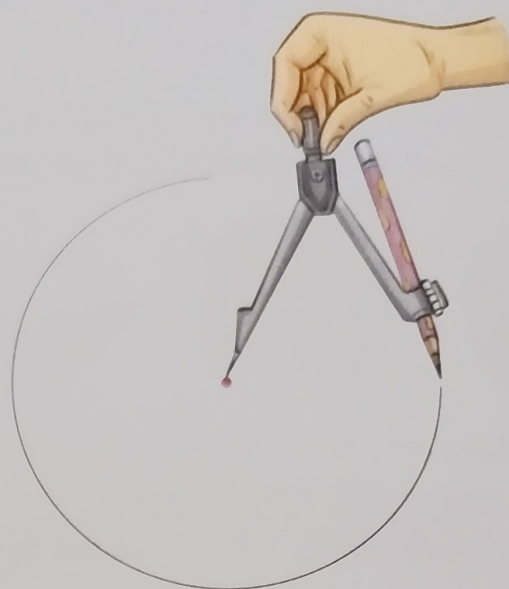
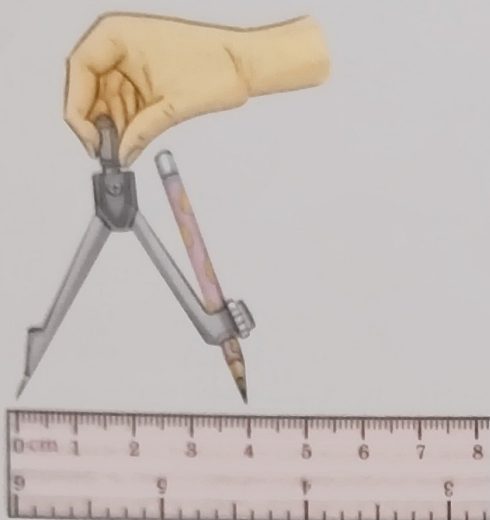
The length of the boundary of a circle is called its **circumference**.

We can see that the circumference of a circle consists of all points that are at the same distance from the centre. This is the length of the radius.

Drawing a Circle

We can draw a circle using a compass. Let us draw a circle of radius 4 cm.

- Fix a sharp pencil in your compass.
- Stretch the arms of the compass so that the metal tip is at the 0 cm mark and the pencil point is at the 4 cm mark.
- Place the metal tip of the compass in the middle of the page and holding it in place, trace a curve by moving the pencil arm around.
- You will obtain a circle.



We can change the radius and draw circles of different sizes. This can be done by opening the arms of the compass so that the pencil tip and the metal tip are at different distances from each other. Place the tips on the scale to measure different distances for the radius.

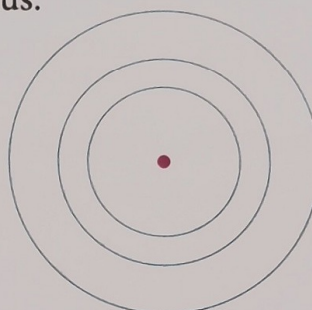
Let's Try! 3

1. Draw a circle with radius 2 cm.
2. Find the diameter of the circle of radius 3 cm.

Concentric Circles

Next draw three circles of radius 1 cm, 1.5 cm, and 2 cm without changing the centre. What do you observe?

Such circles are called **concentric circles**.

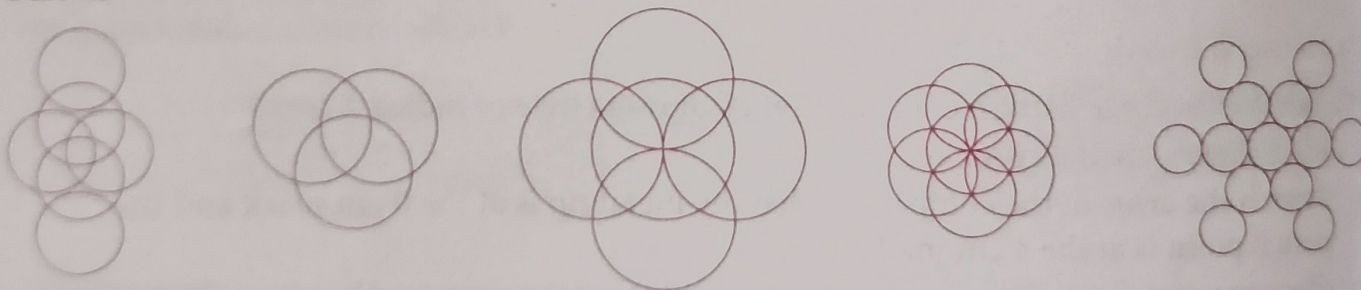


concentric circles



Two or more circles which have the same centre and different radii are called concentric circles.

You can also draw different designs with circles. Have fun colouring the designs given below.



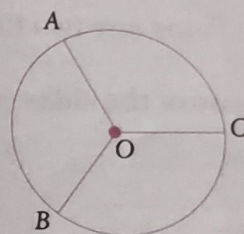
Exercise 11D

Go to MyInspire and attempt questions 1 to 3.

4. Draw a circle of radius 3.5 cm and mark the following.
- Centre
 - Radius
 - Diameter

Application Based Questions

5. Do as directed.
- Mark a point in the middle of the page of your notebook and name it P . With P as the centre draw a circle of radius 5 cm.
 - Mark a point Q anywhere on the circle you have drawn. With Q as centre draw a circle of radius 3 cm.
 - What are the diameters of the circles you drew?
6. Measure the radii OA , OB , and OC in the given figure. What do you observe?



Let's Review



Come here after you have completed the chapter. Colour 1 to 5 😊 to show how well you understand a topic. (5 😊 = Very well; 1 😊 = Not at all)

Basic Geometrical Concepts



Construction of Line Segments



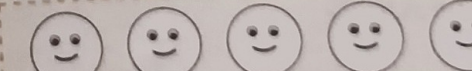
Angles



Open and Closed Figures



Circle



Revision Exercises



Go to MyInspire for



Start Up



1. Define the following.

a. Line

b. Line segment

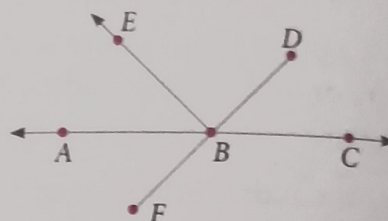
c. Ray

2. Name the following in the given figure.

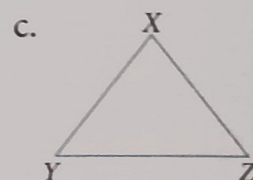
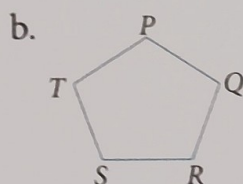
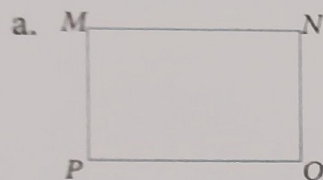
a. Name all the points.

b. Name any three rays.

c. Name any two line segments.



3. Measure the sides of the following figures.



Level Up



1. Draw line segments of the following lengths.

a. $AB = 7.3$ cm

b. $CD = 6.1$ cm

c. $XY = 4.8$ cm

2. Find the diameter of a wheel whose radius is 27 cm.

3. What is the radius of a bangle whose diameter is 8 cm?

4. Draw a circle with centre O and radius 4.8 cm. Draw a radius and a diameter. Name them and write their lengths.

5. Do as directed.

a. Draw a line AB of any length. Cut out a line segment PQ of length 5 cm from line AB .

b. Draw two concentric circles of radii 4 cm and 5 cm with the centre at C. Draw a diameter of the smaller circle and a radius of the bigger one. Name them and write their measurements.